

# Interactive edge awareness network for salient object detection in optical remote sensing images

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## ABSTRACT

Salient object detection in optical remote sensing images (ORSI-SOD) has gained significant attention due to its wide range of applications. However, the complex scenes and large scale variations in ORSI still pose persistent challenges, including deficient boundary representation, severe background clutter, and semantic degradation. Moreover, most existing methods tend to address these challenges independently, lacking an effective mechanism to collaboratively integrate complementary cues. To bridge this gap, we propose the interactive edge awareness network (IEANet) for ORSI-SOD. Specifically, the Interactive Edge Awareness Module (IEAM) integrates a three-branch feature interaction mechanism with an average-max pooling difference strategy, effectively capturing boundary information and adaptively enhancing edge representations without explicit edge supervision. To suppress background clutter and boost multiscale object perception, context attention guidance module (CAGM) employs dual-path attention to highlight salient regions through global-local feature fusion. Moreover, semantic consistency preservation module (SCPM) adopts an attention-driven strategy that alleviates semantic degradation during upsampling and further facilitates the discriminative power of feature decoding in complex environments. Extensive experiments on three benchmark datasets demonstrate the effectiveness of IEANet, which achieves superior performance over 22 state-of-the-art methods on most of the eight evaluation metrics, with the compared methods reporting average S-measure, mean E-measure, and MAE values of 92.53%, 96.11%, and 0.69%, respectively.

## 1. Introduction

Salient object detection (SOD) focuses on identifying regions within an image that exhibit strong visual distinctiveness and attract human attention. These regions correspond to the most perceptually significant areas critical for various applications such as image quality assessment (Zhai & Min, 2020), medical image segmentation (Ramesh et al., 2021), image retargeting (Ding et al., 2011), and video processing (Sharma et al., 2021). Salient object detection in natural scene images (NSI-SOD) (Naosekpan & Sahu, 2022) has been extensively studied and has laid a solid foundation for related tasks. With the development of the field, salient object detection in optical remote sensing images (ORSI-SOD) (Li et al., 2020) has emerged as a derived and rapidly growing research direction, drawing increasing attention from the community. NSI-SOD deals with images characterized by clear targets and distinct edges, while ORSI-SOD is obviously more challenging, as it often involves small target sizes, low foreground-background contrast, and significant inter-

ference from complex ground objects. These challenges make ORSI-SOD a more demanding yet valuable area of research.

Recently, many methods have emerged specifically for ORSI-SOD and fully considered the unique characteristics of remote sensing images. For example, LVNet [21] uses multi-resolution remote sensing images as input to cope with the problem of varying sizes of salient objects. TSTnet (Zhang et al., 2021) aligns topological relationships in hyperspectral images to improve the accuracy of semantic information transfer. In addition, Methods such as MJRBM (Tu et al., 2021), SeaNet (Li et al., 2023c) and ADSTNet (Zhao et al., 2024a) enhance edge information to a certain extent, but most of them still rely on multi-scale perception and edge GT supervision. However, this supervision method can easily cause the model to over-focus on edge details and ignore global features such as texture, color and shape, which in turn affects the accurate recognition of the overall target. To alleviate this problem, AESINet (Zeng et al., 2023) proposed an adaptive edge enhancement method without edge GT, which can dynamically

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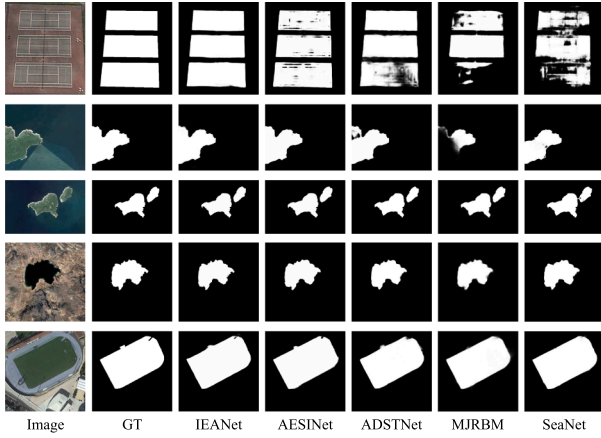


Fig. 1. Saliency maps generated by four comparison methods and our method on ORSI-SOD.

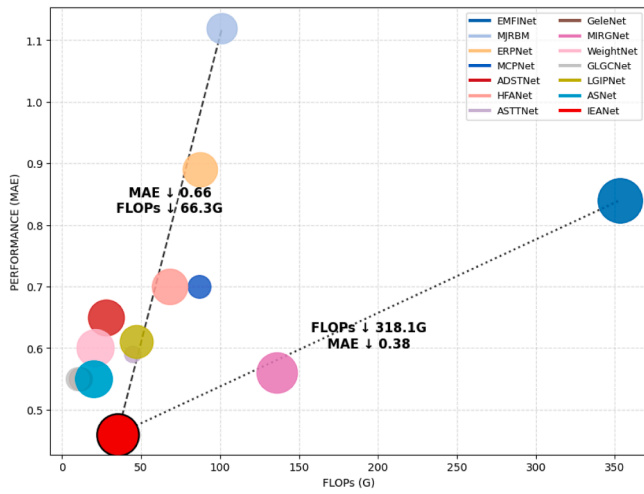


Fig. 2. The performance (MAE) and FLOPs (G) of 15 representative existing ORSI-SOD models and the proposed model are compared, where a larger bubble indicates a larger number of model parameters.

fuse edges and salient features to achieve more natural edge modeling. Nevertheless, edge enhancement may still bias feature learning toward local boundary details while suppressing global semantic cues such as texture, color, and shape. This imbalance limits holistic object understanding and can even distort the overall target structure, as illustrated in Fig. 1, where edge-enhanced features guided by GT unexpectedly undermine structural coherence. Furthermore, when edge enhancement is treated independently or overly emphasized, its coordination with high-level semantic representations becomes insufficient, making it difficult to jointly preserve boundary clarity and global semantic consistency. Consequently, jointly leveraging edge information, global feature modeling, and complementary cue coordination within a unified framework remains challenging in ORSI-SOD Fig. 2.

To address these challenges, we propose a novel interactive edge awareness network (IEANet) that integrates and coordinates multiple complementary cues, including the interactive edge awareness module (IEAM), context attention guidance module (CAGM), and semantic consistency preservation module (SCPM). Specifically, IEAM captures subtle boundary cues by computing discrepancies across three mutually guided perception branches, adaptively emphasizing edge structures without explicit edge supervision. Next, CAGM improves robustness to scale and orientation variations by employing parallel attention mechanisms for global context modeling and local refinement. The coordinated integration of edge-aware and semantic features is achieved by

SCPM in the decoder. By leveraging self-attention to refine feature interactions, semantic degradation is alleviated while boundary sensitivity is preserved, ultimately enabling consistent and noise-resilient feature reconstruction. Through this coordinated feature flow, IEANet effectively integrates complementary edge and semantic cues.

The main contributions of this paper are summarized as follows.

- We propose IEANet to exploit complementary visual representations for accurate salient region perception. Different from existing ORSI-SOD methods that treat heterogeneous feature cues independently, our framework emphasizes explicit coordination among multiple visual clues. Experimental results show the superior performance of our method, outperforming 22 state-of-the-art approaches.
- We develop IEAM that combines a triple-branch interaction mechanism with an average-maximum pooling differential strategy, which adaptively enhances salient object boundaries and captures complex structures through multiscale feature differentials.
- We design CAGM that leverages dual-path spatial and channel attention to obtain both global and local features, complement Transformer representations, and support multiscale salient object detection with fine-grained detail retention.
- We design SCPM that fully utilizes sub-attention mechanisms and focuses on multiscale features to recover degraded semantic information, thereby progressively alleviating semantic degradation in the upsampling process.

The remainder of this paper is organized as follows. Section 2 reviews the related work NSI-SOD and ORSI-SOD. Section 3 presents the proposed methodology in detail. Extensive experimental results and analyses are reported in Section 4. Finally, Section 5 concludes the paper.

## 2. Related work

In this section, we will introduce the classic related research on NSI-SOD, ORSI-SOD and edge-enhanced salient object detection respectively.

### 2.1. SOD in natural scene image

SOD initially emerged within the domain of NSI-SOD analysis, with early developments primarily based on traditional unsupervised methods. These approaches relied on handcrafted features and incorporated techniques such as saliency trees (Liu et al., 2014), random walks (Xia et al., 2019), structured matrix decomposition (Peng et al., 2016), and two-layer sparse graphs (Zhou et al., 2017). While their performance was often constrained by limited representational capacity, these methods laid a foundational theoretical framework and provided key insights that inspired future research directions.

With the advancement of machine learning, supervised and semi-supervised methods were introduced, leading to significant progress in the field. The advent of deep learning, particularly convolutional neural networks (CNNs), marked a turning point in NSI-SOD. Inspired by pioneering architectures such as VGG (Simonyan & Zisserman, 2014) and ResNet (He et al., 2016), many methods adopted encoder-decoder frameworks tailored for pixel-wise prediction tasks. To further improve detection performance, researchers proposed a range of strategies, including global context aggregation (Noman et al., 2024), attention mechanisms (Guo et al., 2022), edge guidance (Zhao et al., 2019), multiscale feature interaction (Liu et al., 2021), and deep supervision (Li et al., 2018). For instance, PoolNet (Liu et al., 2022) demonstrated the benefits of pooling operations for enhanced localization accuracy and detail retention, while EGNet (Zhao et al., 2019) jointly optimized salient and edge information, yielding clearer object boundaries. GC-PANet (Chen et al., 2020) effectively fused low-level details and high-level semantics to generate comprehensive saliency maps. As a result, the adaptation of NSI-SOD strategies to remote sensing imagery requires

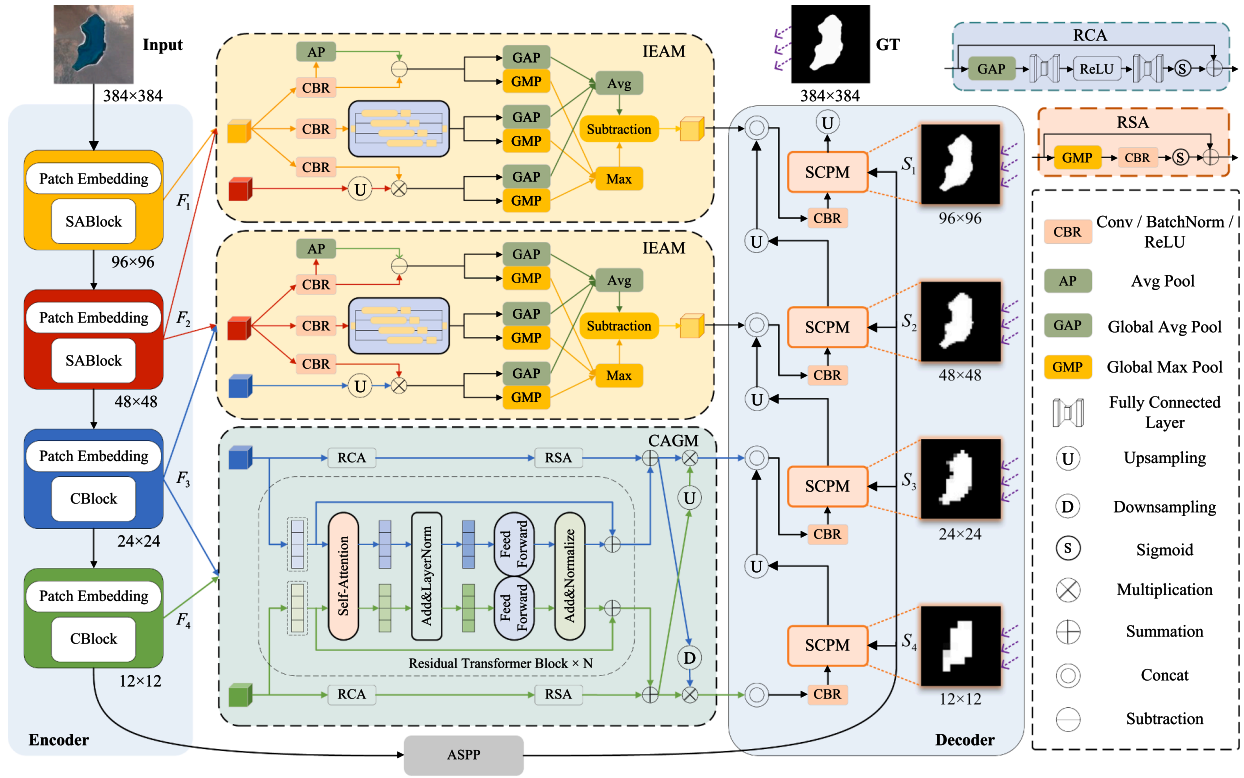


Fig. 3. Overall framework of the proposed IEANet. IEAM and CAGM are employed to refine these embeddings to enhance edge and contextual clues, and SCPCM is employed to mitigate semantic loss during the decoding process.

unified frameworks that jointly incorporate edge awareness, multiscale representation, and contextual reasoning.

## 2.2. SOD in optical remote sensing images

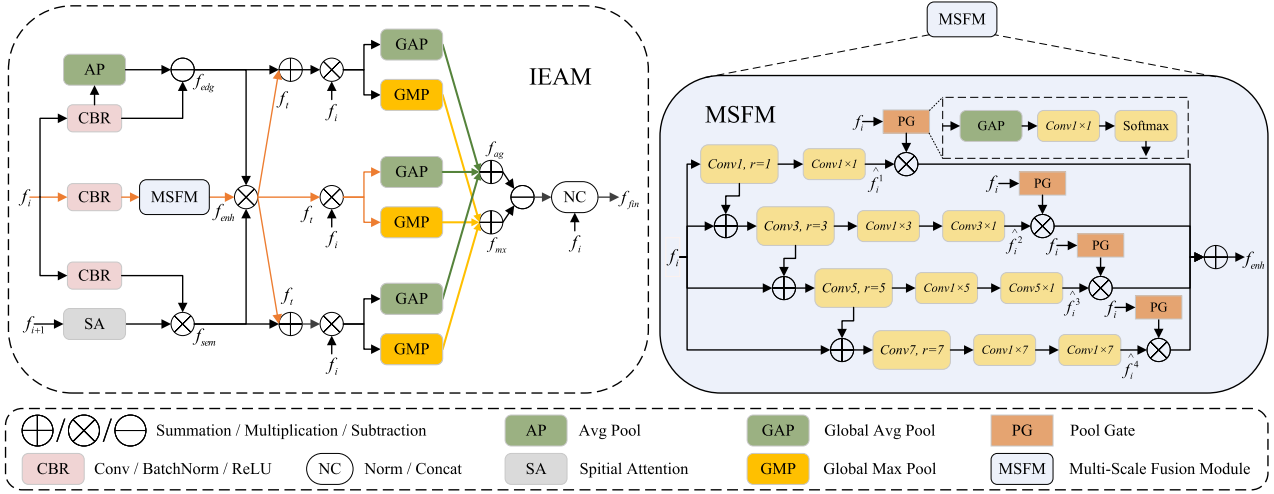
In contrast to the extensively researched NSI-SOD field, ORSI-SOD remains in its early stages, although it has demonstrated accelerated progress in recent years. As a pioneering work, LVNet (Li et al., 2019) tackled scale variation and complex backgrounds by proposing a dual-stream pyramid and nested connection architecture, while also contributing the first public dataset of 800 annotated optical remote sensing images. DAFNet (Zhang et al., 2020) expanded on this foundation by incorporating global contextual awareness and dense attention mechanisms, leading to the creation of the large-scale EORSSD dataset. MJRBM (Tu et al., 2021) introduced a multiscale joint region-boundary model, which jointly optimizes region and edge representations and further enhanced dataset scale with the ORSI-4199 benchmark. However, these CNN-based methods generally suffered from limited receptive fields, restricting their ability to model global dependencies. To address this, GeleNet (Li et al., 2023b) adopted a Transformer-driven approach, combining global context modeling with local refinement through direction-aware convolutions and knowledge transfer modules. Recent developments in the field highlighted a transition toward architectures capable of simultaneously modeling global semantics and local structures in a unified manner.

As scene classification and hyperspectral image analysis continue to evolve, the integration of attention-based mechanisms and multiscale fusion has become a defining strategy for addressing the structural complexity of ORSI-SOD. For example, ACNet (Tang et al., 2021) improved remote sensing scene classification by using spatially rotated image pairs and a parallel attention mechanism to enhance local information extraction. MIFA-Net (Yan et al., 2025) utilized pseudo-boundary masks, image-level category labels, and scribble annotations to highlight detection targets. SpectralFormer (Hong et al., 2021) further advanced

hyperspectral image classification by capturing local spectral sequences through inter-group spectral embedding and adaptive fusion strategies, mitigating information loss during inter-layer propagation. Compared to traditional CNN-based architectures, Transformer-based frameworks better extract global context in remote sensing. PROFILE (Han et al., 2023) proposed a progressive feature interleaving fusion network that interleaved CNNs and Transformers to discern global semantics and spatial details, enhancing the accuracy and integrity of salient target detection under complex scene variations. Nevertheless, directly adopting Transformer-based or hybrid architectures in ORSI-SOD remains challenging, as exhaustive token interactions often introduce redundancy and may compromise the preservation of fine-grained spatial details and adaptability to varying resolutions. In contrast, UniFormer (Li et al., 2023d) integrated convolution and self-attention within a unified architecture, modeling local interactions in shallow layers and global dependencies in deeper layers, thereby achieving efficient yet effective feature representation.

## 2.3. Edge enhancement for SOD

Optical remote sensing images are often characterized by small target sizes and complex edge features, leading to blurred boundaries or foreground-background adhesion in predictions. Enhancing edge feature extraction and generating clear, low-noise boundaries has thus become a key challenge. To address this, many methods introduced explicit edge-related modules to boost edge perception. For example, PoolNet (Liu et al., 2022) employed a pooling-based edge detection branch trained jointly with the main branch, while EG-Net (Zhao et al., 2019) extracted positional and boundary information to improve localization. TSCNet (Li et al., 2023a) leveraged low-level features to render more detailed edges in high-level features. Similarly, CSBNet (He et al., 2024) innovatively integrated category feature-guided structural information extraction and boundary recognition. MEANet (Liang & Luo, 2024) enhanced detection performance



**Fig. 4.** The architecture of the proposed interactive edge awareness module (IEAM). The IEAM adaptively enhances object edges through a three-branch interaction mechanism and pooling-based feature differencing, it automatically captures multiscale edge features and effectively preserves structural details in complex scenes.

significantly by embedding edge information into spatial attention maps and guiding feature fusion with multi-level semantic information. These approaches have demonstrated the effectiveness of edge-specific modules, particularly in improving the delineation of complex object boundaries.

Apart from independently designed edge detection components, attention mechanisms have been introduced in some methods to better focus on informative regions and improve edge perception. ERPNet (Zhou et al., 2022) introduced an edge position attention unit. ESGNet (Gong et al., 2023) adopted a two-stage strategy to enhance edge and skeleton cues, and DAFNet (Zhang et al., 2020) fused global attention with edge supervision to refine object contours. These approaches suggested that rather than isolating edge features, integrating them with semantic and enhancement branches may lead to more precise localization. ITSD (Zhou et al., 2020) exemplified this idea through a lightweight decoder that explored the interaction between salient regions and contours. From a broader perspective, object boundaries not only delineate foreground and background, but also serve as critical structural constraints that guide semantic aggregation and suppress background interference. This insight underscores the necessity of modeling edge information as a core element within feature interaction and representation learning, rather than treating it as an isolated auxiliary signal.

### 3. Method

In this section, we first introduce on the overall architecture of the proposed IEANet. As shown in the Fig. 3, we use UniFormer (Li et al., 2023d) as the encoder of the model. Subsequently, we describe the three key module structures and the motivations in detail in Sec.III-B, Sec.III-C, Sec.III-D. Finally, the loss functions employed in our model are presented.

#### 3.1. Overview

The proposed IEANet architecture is illustrated in the Fig. 3, which comprises an encoder, several intermediate modules, and a decoder. The input size of our IEANet is  $3 \times 384 \times 384$ . We use UniFormer as the encoder backbone, which is organized into four stages with distinct computational blocks. Convolutional block (CBlock) are adopted in stages 1 and 2 to model local representations, while self-attention block (SABlock) are utilized in stages 3 and 4 to capture global contextual information. Each stage outputs a feature map, collectively forming a four-

level hierarchical representation, denoted as  $F_t \in \mathbb{R}^{c_t \times h_t \times w_t}$  ( $t = 1, 2, 3, 4$ ), where  $h_t$  and  $w_t$  are  $(384/2^{t+1})$ , and  $c_t \in \{64, 128, 320, 512\}$ , which are subsequently fed into four specialized decoders. Unlike existing methods that typically arrange attention mechanisms and edge enhancement modules in a sequential or parallel manner, IEANet is designed with an explicit collaborative interaction among IEAM, CAGM, and SCPM. Specifically, IEAM does not merely enhance edge cues or semantic features independently; instead, it interactively refines both through mutual guidance between fine-grained spatial details and high-level semantic representations. This joint optimization enables the model to preserve structural integrity while improving contour clarity, rather than treating boundary enhancement as an isolated preprocessing step. Building upon this interaction, CAGM further aggregates multi-level deep features by integrating contextual information across different layers, facilitating a bidirectional information flow between global semantics and local details. This design contrasts with conventional multi-scale fusion strategies that pass information in a unidirectional manner, and allows salient features to be progressively calibrated across scales. Finally, SCPM enforces semantic consistency during decoding by coordinating multiscale attention within each decoder branch, alleviating semantic degradation caused by upsampling. Through this tightly coupled IEAM-CAGM-SCPM interaction pipeline, IEANet achieves collaborative feature integration rather than a simple composition of independent modules.

#### 3.2. Interactive edge awareness module

The ORSI-SOD architecture relies heavily on detailed feature extraction, particularly for capturing the complex edges of salient objects. Since such fine-grained information is primarily embedded in the shallow outputs of the encoder, we introduce the IEAM module as shown in Fig. 4 to effectively exploit these features. Assume that the two inputs are  $f_i$  and  $f_{i+1}$ , where  $f_{i+1}$  represents the higher-level feature. The multi-branch parallel structure improves edge perception by enabling specialization in complementary features, such as fine spatial details and contextual semantics. Accordingly,  $f_i$  is divided into three branches which focus on feature enhancement, edge information, and semantic information respectively. In each branch, we use CBR (Li et al., 2024a) to alleviate the issues of unstable gradient flow and internal covariate shift commonly encountered.

Single-scale feature extraction fails to discern diverse object sizes and contextual information, which limits the ability to adapt to varying scales and details (Singh & Davis, 2018). Therefore, we add a multiscale



fusion module to the feature enhancement branch. The architecture is decomposed into four parallel branches with gradually expanded receptive fields ( $r = 1, 3, 5, 7$ ), which can capture both local texture and global context patterns. Each branch further contains asymmetric decomposed convolutions ( $1 \times 3$  and  $3 \times 1$ ,  $1 \times 5$  and  $5 \times 1$ ,  $1 \times 7$  and  $7 \times 1$ ) to reduce computational cost and maintain spatial sensitivity. The process is as follows

$$\begin{aligned}\hat{f}_i^1 &= \text{Conv}1 \times 1(\text{Conv}1(f_i^1)) \\ \hat{f}_i^2 &= \text{Conv}1 \times 3(\text{Conv}3 \times 1(\text{Conv}3(f_i^2 + \text{Conv}1(f_i^1)))) \\ \hat{f}_i^3 &= \text{Conv}1 \times 5(\text{Conv}5 \times 1(\text{Conv}5(f_i^3 + \text{Conv}3(f_i^2)))) \\ \hat{f}_i^4 &= \text{Conv}1 \times 7(\text{Conv}7 \times 1(\text{Conv}7(f_i^4 + \text{Conv}5(f_i^3))))\end{aligned}\quad (1)$$

where  $\text{Conv}_i$  represents the branch-specific convolution with receptive field  $r = i$ , and  $\text{Conv}_{i \times i}$  denotes the convolution kernel size. In addition, we introduce a novel pooling gate mechanism to adaptively regulate the contributions of different branches according to cross-channel statistics. Global average pooling is first employed to aggregate channel-wise responses, followed by a softmax-based gating function that generates normalized and competitive branch weights. The process is as follows

$$f_{pg}^n = \text{Softmax}(\text{Conv}1 \times 1(\text{GAP}(\hat{f}_i^n))), \quad n = 1, 2, 3, 4 \quad (2)$$

where Softmax represents the softmax function, GAP represents global average pooling, and  $f_{pg}^n$  represents the features generated by gating. Finally, the four branches are summed to obtain enhanced features.

$$f_{enh} = \sum_{n=1}^4 \hat{f}_i^n \times f_{pg}^n \quad (3)$$

Pooling-subtraction is employed in the edge branch to enhance edge features by suppressing background noise. By subtracting an average-pooled version from the original features, this operation functions as a spatial high-pass filter that preserves boundary transitions while eliminating redundant regions. The process is described as

$$f_{edg} = f_i \ominus \text{AP}(f_i) \quad (4)$$

where AP represents the average pooling. To enhance semantic representation, we apply the SA mechanism to high-level features to capture long-range context, then multiply the result with  $f_i$  to form the semantic branch, which enables dynamic feature recalibration for more discriminative representations. The process is described as

$$f_{sem} = f_i \times \text{SA}(f_{i+1}) \quad (5)$$

where SA represents the spatial attention. Traditional edge detection struggles to balance fine detail preservation and noise suppression, leading to inaccurate boundaries and difficulty distinguishing object edges from background clutter due to limited context. To address these issues, we propose an interaction strategy that fuses multi-level features with semantic cues to enhance boundary clarity and segmentation accuracy. Specifically, pixel-wise multiplication of the three branch features yields an interaction-aware feature map  $f_i$ , which is then added back to each branch and fused with the original residual feature  $f_i$  to enhance cross-branch consistency and effectively strengthen edge perception through coordinated feature refinement. Furthermore, inspired by the edge-aware design in AESINet (Zeng et al., 2023), which exploits the difference between global maximum and average pooling to extract edge features adaptively, we apply both pooling operations to each branch independently and sum the results to generate feature maps  $f_{mx}$  and  $f_{ag}$ . The final feature representation is

$$f_{fin} = \text{Concat}(f_i, \text{Norm}(f_{mx} - f_{ag})) \quad (6)$$

### 3.3. Context attention guidance module

In ORSI-SOD, the presence of complex scene layouts, diverse object scales, and significant intra-class variation poses considerable challenges for accurate feature extraction and object delineation. Salient objects in such images are typically characterized by irregular shapes and

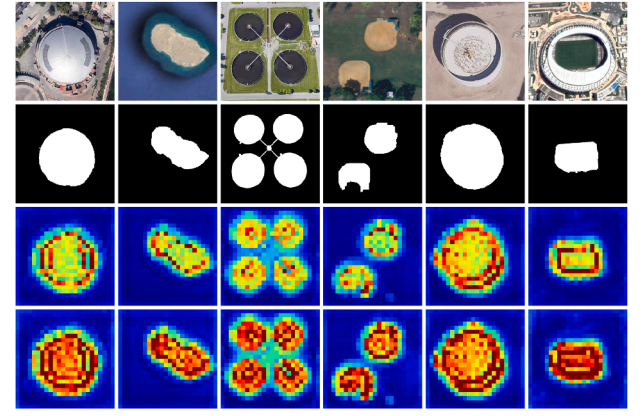


Fig. 5. Heat map analysis of Transformer block. The first row shows the selected several representative original input images from the EORSSD dataset. The second row shows the ground truth. The third row shows the responded maps without Transformer block. The fourth row shows the responded maps of the entire model.

indistinct boundaries, and are often embedded in heterogeneous backgrounds that share similar spectral or textural properties with the foreground. Furthermore, traditional convolutional-based methods (Li et al., 2019) are inherently limited by their constrained receptive fields, which hampers their ability to extract long-range context and often leads to the loss of fine-grained spatial details needed for precise localization. Due to their limited capacity to jointly model global and local features, these methods tend to produce fragmented objects and inaccurate boundaries (Fig. 6).

To address these limitations, we introduce the CAGM, which aims to enhance feature representation by integrating multi-dimensional contextual information. In designing this module, we incorporated residual channel attention (RCA) and residual spatial attention (RSA) mechanisms in parallel, inspired by CBAM (Woo et al., 2018), to adaptively recalibrate feature responses along both channel and spatial dimensions. The RCA mechanism models inter-channel dependencies by first applying global average pooling to aggregate spatial information into a compact descriptor, and then processing it through a two-layer fully connected network with a ReLU activation to capture nonlinear channel-wise relationships. The resulting output, modulated by a sigmoid function, is used to reweight the original feature map. The process is described as

$$f_{rea} = f_i + \alpha(\text{CBR}(\text{GMP}(f_i))) \quad (7)$$

where  $\alpha$  denotes the sigmoid function, CBR represents a sequence of convolution, batch normalization, and ReLU, and GMP denotes global max pooling. Concurrently, the RSA mechanism emphasizes spatial dependencies by applying global max pooling across the channel axis to highlight salient spatial regions, followed by a convolutional block with batch normalization and ReLU activation to model complex spatial interactions. Subsequently, a sigmoid activation function generates the spatial attention map for spatial-wise reweighting, and a residual connection is introduced to preserve the initial spatial structure and facilitate stable refinement. Its structure is as follows

$$f_{rsa} = f_i + \alpha(\text{FC}(\text{ReLU}(\text{FC}(\text{GAP}(f_i)))) \quad (8)$$

where GAP denotes global average pooling, FC denote fully connected layer. As illustrated in Fig. 5, the heatmap visualization demonstrates the effectiveness of the Transformer block in enhancing feature representation. A comparison between the third and second rows clearly shows that the Transformer greatly expands the activation coverage of salient objects. This evidences that the proposed module not only strengthens the representational capacity of features, but also substantially enhances the ability to accurately locate and interpret key regions, particularly in complex and cluttered remote sensing scenes.

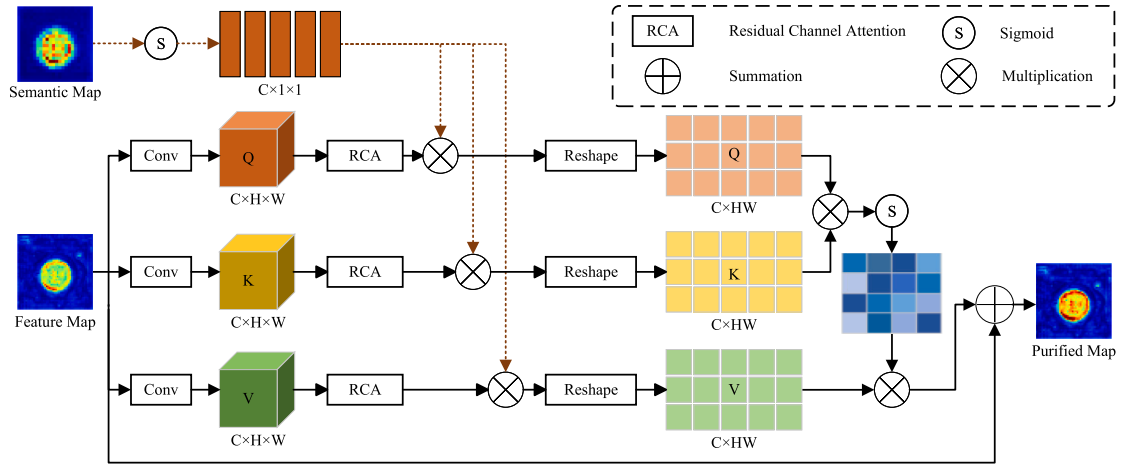


Fig. 6. The architecture of the proposed semantic consistency preservation module (SCPM) dynamically corrects semantic degradation during upsampling via an attention mechanism. It mitigates information loss and selectively enhances multiscale features in complex scenes.

### 3.4. Semantic consistency preservation module

Maintaining semantic consistency is crucial in salient object detection and image segmentation tasks. However, deep learning models may suffer from semantic information loss or inconsistency during multi-layer feature fusion. First, the upsampling process progressively dilutes the semantic information of deep features, thereby weakening semantic representations and ultimately causing edge fragmentation and internal semantic holes in salient objects. Second, there are semantic differences between feature maps of different scales, and simple fusion methods may lead to semantic confusion. To address this issue, we incorporate an ASPP (Chen et al., 2019) into the highest-level feature maps and further integrate it into the decoder at each layer, in order to enhance semantic consistency across the network and effectively mitigate semantic loss and ambiguity during multiscale feature fusion.

Specifically, given an input feature map  $f \in \mathbb{R}^{C \times H \times W}$ , we first extract global semantic dependencies across channels by applying global average pooling, then refine these dependencies through two fully connected layers separated by a ReLU activation, as described by the following

$$f_c = FC \cdot \text{ReLU}(FC \cdot \text{GAP}(f)) \quad (9)$$

Next, to enhance multiscale contextual information within the deep features and further strengthen semantic representation, we apply an ASPP to get the semantic map. The obtained semantic map is modulated by the output of the fully connected layer after a sigmoid activation, enabling adaptive enhancement of semantically important regions through element-wise multiplication. The process is represented as

$$f_s = \alpha(f_{aspp}) \times f_c \quad (10)$$

where  $\alpha$  represents the sigmoid function. To further refine semantic consistency across scales and model long-range dependencies, we introduce a lightweight self-attention mechanism. Specifically, we generate the query (Q), key (K), and value (V) projections as follows:

$$\begin{cases} Q = W_q \times f_s \\ K = W_k \times f_s \\ V = W_v \times f_s \end{cases} \quad (11)$$

where  $W_q, W_k, W_v$  are implemented as  $1 \times 1$  convolutions. The attention map between  $Q$  and  $K$  is normalized via a sigmoid function, which enables flexible modeling of semantic correlations without incurring quadratic complexity. Furthermore, a residual connection is introduced to preserve the original feature representations and stabilize information propagation. This design allows the module to capture long-range

semantic dependencies while maintaining low computational cost, making it well suited for semantic refinement during upsampling.

$$f_{out} = \alpha(Q \cdot K^T) \cdot V + f \quad (12)$$

### 3.5. Loss function

We use intersection over union (IOU) (Rahman & Wang, 2016) loss and binary cross entropy (BCE) (Ruby et al., 2020) loss to train the proposed network. Among them, IoU loss focuses more on the accuracy of the model on the target location and shape, and it optimizes this ability by minimizing the IoU difference between the predicted image and the real image. The process is as follows

$$L_{IoU} = 1 - \frac{\sum_i p_i g_i}{\sum_i p_i + \sum_i g_i - \sum_i p_i g_i} \quad (13)$$

where  $p_i$  represents the value of the  $i$ -th pixel in the predicted image, and  $g_i$  represents the corresponding truth value. Unlike IoU, BCE makes the gradient change smoother, which is very common in binary classification tasks and can effectively distinguish the foreground and background. The process is outlined as follows:

$$L_{BCE} = -\frac{1}{N} \sum_{i=1}^N [g_i \log(q_i) + (1 - g_i) \log(1 - q_i)] \quad (14)$$

where  $q_i$  represents the predicted probability of the  $i$ -th pixel belonging to the foreground, and  $g_i$  represents the corresponding ground truth label,  $N$  denotes the total number of pixels in the image. The overall training objective is defined as a simple combination of the two loss terms:

$$L_{total} = L_{BCE} + L_{IoU} \quad (15)$$

where both loss components are equally weighted. In general, such a combination allows the model to take into account both target classification and target localization, thereby achieving better results in training.

## 4. Experiments

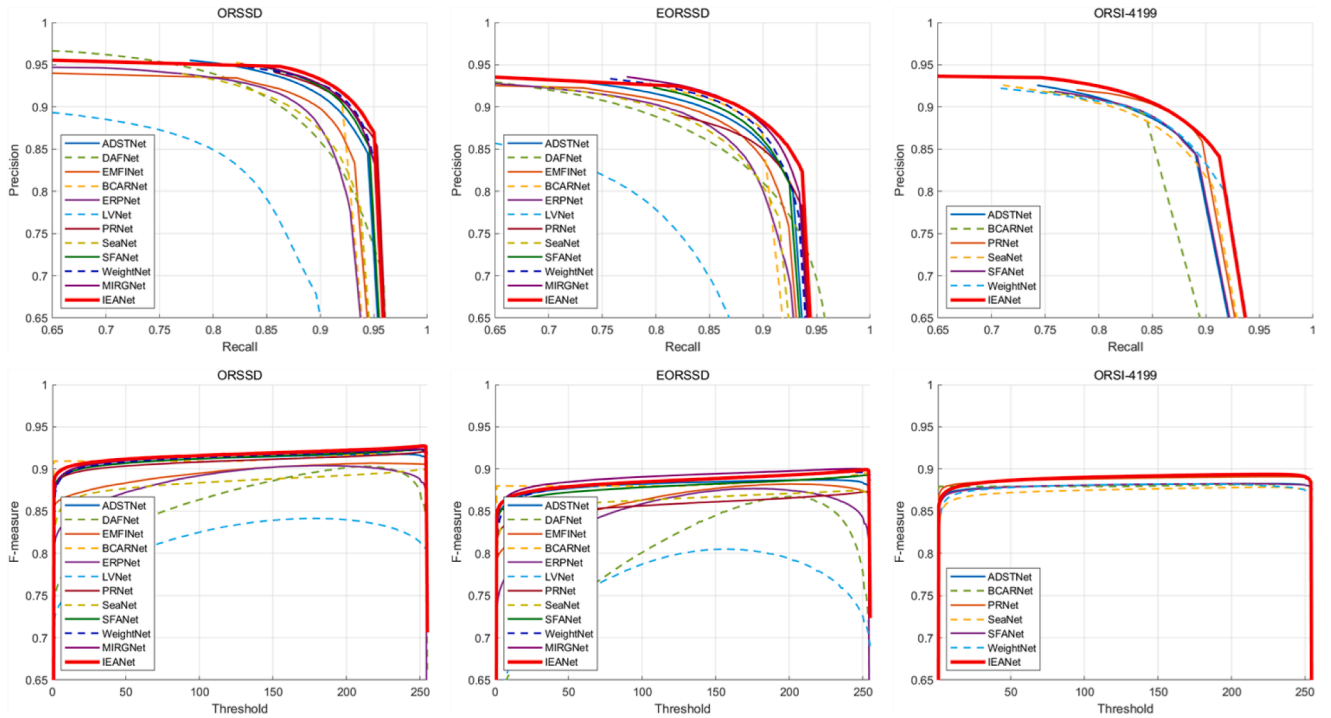
### 4.1. Datasets

We evaluated the proposed method on three challenging ORSI-SOD datasets. The ORSSD dataset (Li et al., 2019) is the first public dataset for ORSI-SOD. Its data comes from Google Earth and some existing ORSI-SOD datasets. It contains 800 images with different diversity of salient objects and cluttered backgrounds, of which

**Table 1**

Quantitative comparisons between the proposed method and 22 state-of-the-art methods.  $\uparrow$  and  $\downarrow$  mean that the higher and lower is better. The top three results are marked in bold, italic, underline, respectively. Among them,  $\dagger$  stands for VGG-based ORSI-SOD method,  $\ddagger$  stands for ResNet-based ORSI-SOD method,  $\S$  stands for Hybrid-based ORSI-SOD method, and  $\P$  stands for Transformer-based ORSI-SOD method.

| Model                                   | Year | ORSSD                       |                                    |                                   |                             |                                    |                                   |                       |                  |  |  | EORSSD                      |                                    |                                   |                             |                                    |                                   |                       |                  |  |  |
|-----------------------------------------|------|-----------------------------|------------------------------------|-----------------------------------|-----------------------------|------------------------------------|-----------------------------------|-----------------------|------------------|--|--|-----------------------------|------------------------------------|-----------------------------------|-----------------------------|------------------------------------|-----------------------------------|-----------------------|------------------|--|--|
|                                         |      | $F_{\beta}^{\max} \uparrow$ | $F_{\beta}^{\text{mean}} \uparrow$ | $F_{\beta}^{\text{adp}} \uparrow$ | $E_{\zeta}^{\max} \uparrow$ | $E_{\zeta}^{\text{mean}} \uparrow$ | $E_{\zeta}^{\text{adp}} \uparrow$ | $S_{\alpha} \uparrow$ | MAE $\downarrow$ |  |  | $F_{\beta}^{\max} \uparrow$ | $F_{\beta}^{\text{mean}} \uparrow$ | $F_{\beta}^{\text{adp}} \uparrow$ | $E_{\zeta}^{\max} \uparrow$ | $E_{\zeta}^{\text{mean}} \uparrow$ | $E_{\zeta}^{\text{adp}} \uparrow$ | $S_{\alpha} \uparrow$ | MAE $\downarrow$ |  |  |
| LVNet $\dagger$ (Li et al., 2019)       | 2019 | 0.8263                      | 0.7995                             | 0.7506                            | 0.9456                      | 0.9259                             | 0.9195                            | 0.8815                | 0.0207           |  |  | 0.7794                      | 0.7328                             | 0.6284                            | 0.9254                      | 0.8801                             | 0.8445                            | 0.8630                | 0.0146           |  |  |
| DAFNet $\dagger$ (Zhang et al., 2020)   | 2020 | 0.8928                      | 0.8511                             | 0.7876                            | 0.9771                      | 0.9539                             | 0.9369                            | 0.9191                | 0.0113           |  |  | 0.8614                      | 0.7845                             | 0.6427                            | 0.9661                      | 0.9291                             | 0.8446                            | 0.9166                | 0.0060           |  |  |
| MJRBM $\dagger$ (Tu et al., 2021)       | 2021 | 0.8854                      | 0.8557                             | 0.7982                            | 0.9733                      | 0.9478                             | 0.9410                            | 0.9182                | 0.0138           |  |  | 0.8289                      | 0.7864                             | 0.6967                            | 0.9461                      | 0.9075                             | 0.8947                            | 0.8971                | 0.0112           |  |  |
| EMFINet $\dagger$ (Wang et al., 2022b)  | 2022 | 0.9002                      | 0.8856                             | 0.8617                            | 0.9737                      | 0.9671                             | 0.9663                            | 0.9366                | 0.0109           |  |  | 0.8720                      | 0.8486                             | 0.7984                            | 0.9711                      | 0.9604                             | 0.9501                            | 0.9290                | 0.0084           |  |  |
| ERPNet $\dagger$ (Zhou et al., 2022)    | 2022 | 0.8974                      | 0.8745                             | 0.8356                            | 0.9710                      | 0.9566                             | 0.9520                            | 0.9254                | 0.0135           |  |  | 0.8632                      | 0.8304                             | 0.7554                            | 0.9603                      | 0.9401                             | 0.9228                            | 0.9210                | 0.0089           |  |  |
| HFANet $\ddagger$ (Wang et al., 2022a)  | 2022 | 0.9112                      | 0.8981                             | 0.8819                            | 0.9770                      | 0.9712                             | 0.9722                            | 0.9399                | 0.0092           |  |  | 0.8876                      | 0.8681                             | 0.8365                            | 0.9740                      | 0.9679                             | 0.9644                            | <u>0.9380</u>         | 0.0070           |  |  |
| SeaNet $\ddagger$ (Li et al., 2023c)    | 2023 | 0.9125                      | 0.8954                             | 0.8801                            | 0.9767                      | 0.9722                             | 0.9670                            | 0.9259                | 0.0105           |  |  | 0.8917                      | 0.8785                             | 0.8561                            | 0.9710                      | 0.9651                             | 0.9602                            | 0.9218                | 0.0073           |  |  |
| GLGCNet $\ddagger$ (Bai et al., 2023)   | 2023 | 0.9236                      | 0.9054                             | 0.8931                            | 0.9864                      | <u>0.9820</u>                      | <u>0.9800</u>                     | <u>0.9488</u>         | <u>0.0071</u>    |  |  | 0.8924                      | 0.8714                             | 0.8499                            | <u>0.9803</u>               | <u>0.9751</u>                      | 0.9701                            | 0.9375                | 0.0055           |  |  |
| ASTTNet $\P$ (Gao et al., 2023)         | 2023 | 0.9061                      | 0.8811                             | 0.8456                            | 0.9795                      | 0.9699                             | 0.9587                            | 0.9348                | 0.0094           |  |  | 0.8741                      | 0.8297                             | 0.7534                            | 0.9757                      | 0.9584                             | 0.9247                            | 0.9253                | 0.0059           |  |  |
| GeleNet $\ddagger$ (Li et al., 2023b)   | 2023 | 0.9203                      | <u>0.9093</u>                      | <u>0.9038</u>                     | 0.9829                      | 0.9779                             | <u>0.9805</u>                     | 0.9410                | 0.0080           |  |  | 0.8774                      | 0.8649                             | 0.8528                            | 0.9794                      | <u>0.9752</u>                      | <u>0.9713</u>                     | 0.9259                | 0.0055           |  |  |
| ADSTNet $\ddagger$ (Zhao et al., 2024a) | 2024 | 0.9124                      | 0.9042                             | 0.8979                            | 0.9807                      | 0.9740                             | 0.9785                            | 0.9379                | 0.0086           |  |  | 0.8804                      | 0.8716                             | 0.8532                            | 0.9769                      | 0.9709                             | 0.9681                            | 0.9311                | 0.0065           |  |  |
| MCPNet $\ddagger$ (Huang et al., 2024)  | 2024 | 0.9135                      | 0.8979                             | 0.8726                            | 0.9807                      | 0.9736                             | 0.9724                            | 0.9433                | 0.0090           |  |  | 0.8868                      | 0.8620                             | 0.8077                            | 0.9765                      | 0.9657                             | 0.9515                            | 0.9373                | 0.0070           |  |  |
| WeightNet $\ddagger$ (Di et al., 2024)  | 2024 | 0.9200                      | 0.9048                             | 0.8911                            | 0.9807                      | 0.9766                             | 0.9719                            | 0.9478                | 0.0082           |  |  | 0.8885                      | 0.8731                             | 0.8409                            | 0.9782                      | 0.9729                             | 0.9631                            | <u>0.9380</u>         | 0.0060           |  |  |
| MIRGNet $\ddagger$ (Zhao et al., 2024b) | 2024 | 0.9192                      | <u>0.9082</u>                      | 0.9001                            | 0.9812                      | 0.9771                             | 0.9789                            | 0.9455                | 0.0086           |  |  | <u>0.8930</u>               | <u>0.8813</u>                      | 0.8588                            | 0.9789                      | 0.9742                             | 0.9679                            | <u>0.9383</u>         | 0.0056           |  |  |
| SFANet $\ddagger$ (Quan et al., 2024)   | 2024 | 0.9192                      | 0.9063                             | 0.8984                            | 0.9830                      | 0.9789                             | 0.9765                            | 0.9453                | <u>0.0077</u>    |  |  | 0.8833                      | 0.8680                             | 0.8492                            | 0.9769                      | 0.9726                             | 0.9669                            | 0.9349                | 0.0058           |  |  |
| GCRNet $\ddagger$ (Li et al., 2024b)    | 2024 | 0.9170                      | 0.9073                             | 0.8990                            | 0.9801                      | 0.9762                             | 0.9758                            | 0.9440                | 0.0084           |  |  | 0.8923                      | 0.8710                             | 0.8448                            | 0.9688                      | 0.9738                             | 0.9697                            | 0.9346                | 0.0069           |  |  |
| ASNet $\ddagger$ (Yan et al., 2024)     | 2024 | 0.9172                      | 0.9072                             | 0.8986                            | 0.9803                      | 0.9764                             | 0.9795                            | 0.9441                | 0.0081           |  |  | <u>0.8959</u>               | 0.8770                             | <u>0.8672</u>                     | 0.9783                      | 0.9745                             | <u>0.9748</u>                     | 0.9345                | 0.0055           |  |  |
| PRNet $\ddagger$ (Gu et al., 2024)      | 2024 | 0.9153                      | 0.9013                             | 0.8915                            | <u>0.9866</u>               | <u>0.9824</u>                      | 0.9779                            | 0.9468                | <b>0.0068</b>    |  |  | 0.8707                      | 0.8510                             | 0.8282                            | <u>0.9809</u>               | 0.9732                             | 0.9625                            | 0.9296                | <u>0.0053</u>    |  |  |
| SOLNet $\ddagger$ (Gu et al., 2025)     | 2025 | 0.9012                      | 0.8946                             | 0.8925                            | 0.9734                      | 0.9689                             | 0.9697                            | 0.9284                | 0.111            |  |  | 0.8609                      | 0.8513                             | 0.8392                            | 0.9623                      | 0.9577                             | 0.9596                            | 0.9171                | 0.0078           |  |  |
| EANet $\ddagger$ (Gu et al., 2025)      | 2025 | 0.9180                      | -                                  | -                                 | 0.9759                      | -                                  | -                                 | 0.9310                | 0.113            |  |  | 0.8974                      | -                                  | -                                 | 0.9740                      | -                                  | -                                 | 0.9171                | 0.0073           |  |  |
| BCARNet $\ddagger$ (Gu et al., 2025)    | 2025 | 0.9196                      | 0.9070                             | <b>0.9073</b>                     | 0.9833                      | 0.9799                             | 0.9796                            | 0.9465                | <u>0.0071</u>    |  |  | 0.8871                      | 0.8736                             | <b>0.8740</b>                     | 0.9761                      | 0.9728                             | 0.9634                            | 0.9361                | <u>0.0054</u>    |  |  |
| LGIPNet $\ddagger$ (Sun et al., 2025)   | 2025 | <u>0.9232</u>               | 0.9060                             | 0.8933                            | 0.9862                      | 0.9808                             | 0.9794                            | <u>0.9494</u>         | 0.0076           |  |  | 0.8859                      | 0.8729                             | 0.8604                            | 0.9760                      | 0.9705                             | 0.9707                            | 0.9347                | 0.0061           |  |  |
| IEANet $\P$                             | -    | <b>0.9255</b>               | <b>0.9107</b>                      | <u>0.9003</u>                     | <b>0.9872</b>               | <b>0.9832</b>                      | <b>0.9827</b>                     | <u>0.9514</u>         | <b>0.0068</b>    |  |  | <b>0.8970</b>               | <b>0.8822</b>                      | <u>0.8646</u>                     | <b>0.9847</b>               | <b>0.9793</b>                      | <b>0.9759</b>                     | <b>0.9419</b>         | <b>0.0046</b>    |  |  |

**Fig. 7.** Comparisons with other methods in terms of PR and F-measure curves on three benchmark datasets.

600 images are used for training and 200 images are used for testing. The EORSSD dataset (Zhang et al., 2020) expands the original ORSSD dataset to 2000 images with richer scenes, of which 1400 images are used for training and 600 images are used for testing. The ORSI-4199 (Tu et al., 2021) dataset is the largest and most challenging dataset for ORSI-SOD, containing 4199 images, of which 2000 images are used for training and 2199 images are used for testing Fig. 9.

#### 4.2. Evaluation metrics

We evaluated the performance of our proposed model based on eight commonly used metrics. The mean absolute error (MAE) Perazzi et al. (2012) is the average of the absolute errors between the predicted value and the true value, which is used to measure the prediction accuracy of the model. We use S-measure (Fan et al., 2017) ( $S_{\alpha}$ ,  $\alpha = 0.5$ ) to evaluate the performance of the saliency detection model, which

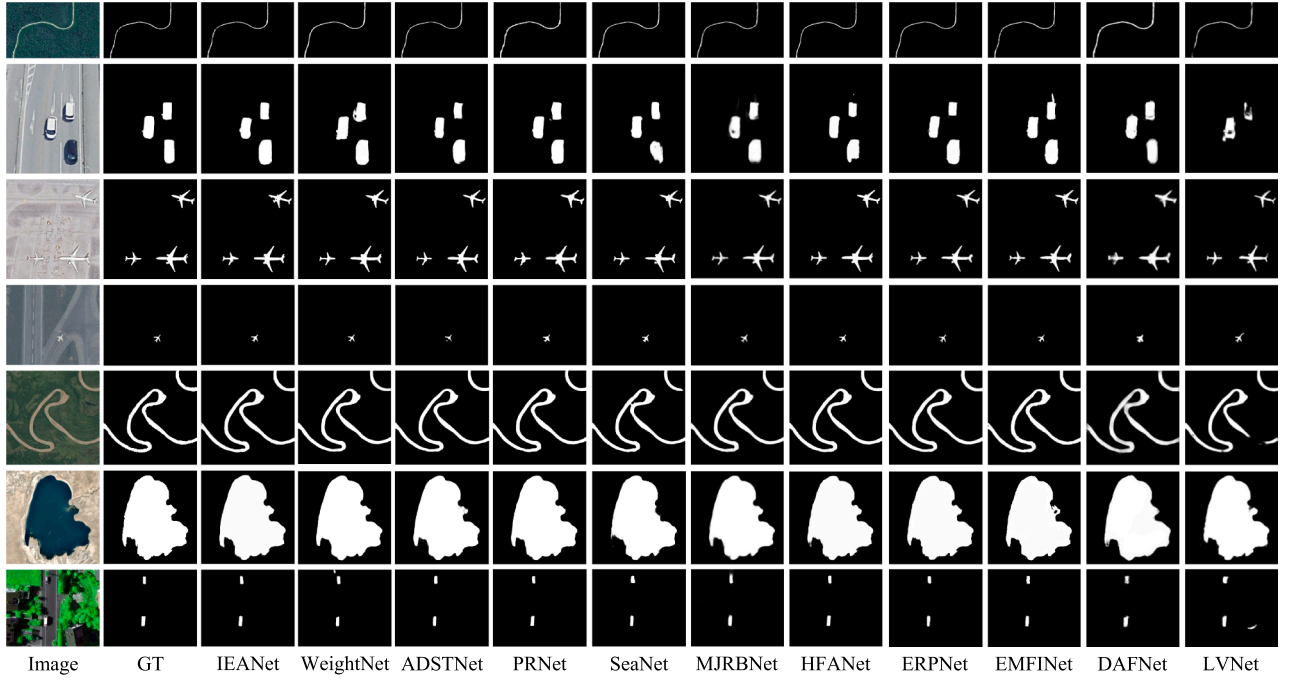


Fig. 8. Qualitative comparison with 10 state-of-the-art methods on representative scenes in ORSI-SODs.

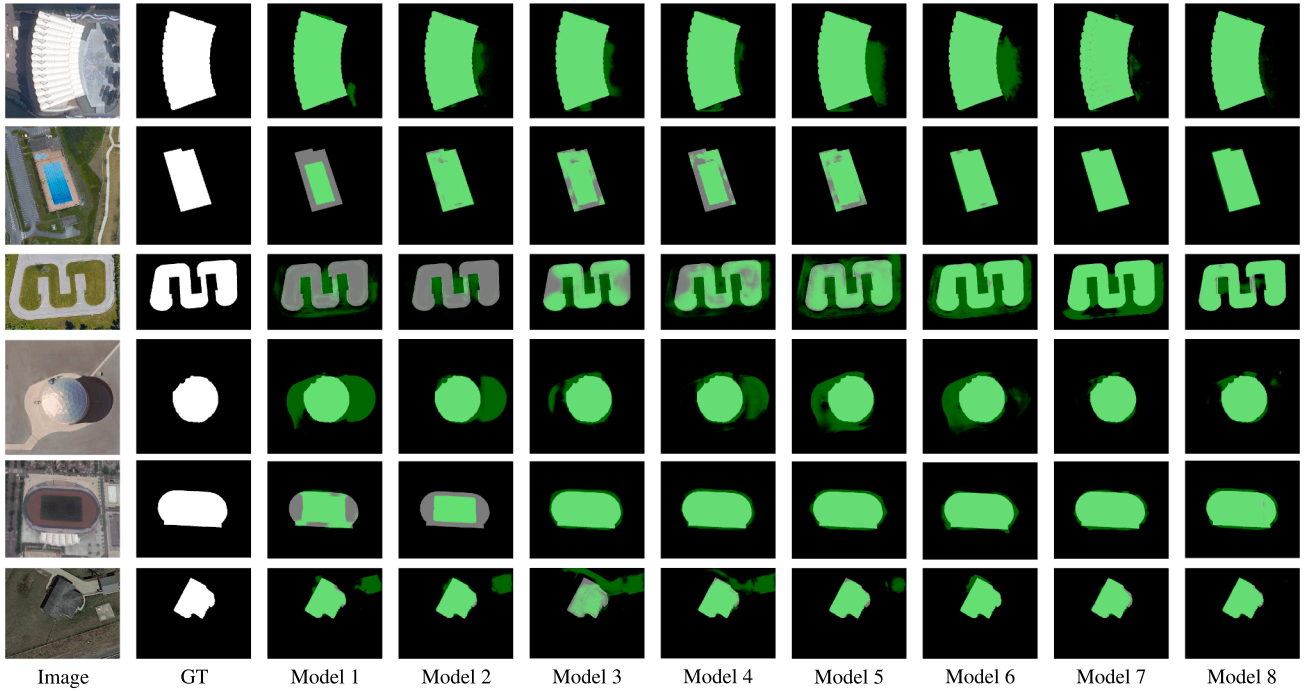


Fig. 9. Visualization of different module composition predictions, where models 1-8 represent the combinations with IDs 1-8 in Table 3.

comprehensively considers regional structure and object perception. Also, we report the maximum, average, and adaptive values of F-measure (Achanta et al., 2009) ( $F_{\beta}^{max}$ ,  $F_{\beta}^{mean}$ ,  $F_{\beta}^{adp}$ ) to comprehensively evaluate the accuracy and recall of the model. Similarly, we also provide the maximum, average, and adaptive E-measure (Fan et al., 2018) ( $E_{\xi}^{max}$ ,  $E_{\xi}^{mean}$ ,  $E_{\xi}^{adp}$ ) to evaluate the performance of the saliency detection model.

#### 4.3. Implementation details

During training and testing, all experiments are conducted on a single NVIDIA RTX 3090 GPU. To enhance the robustness of IEANet and prevent overfitting, data augmentation is performed by rotating the training images by 90°, 180°, and 270°, as well as applying mirror flipping followed by the same rotations. Through this strategy, the training



**Table 2**

quantitative comparison of the weightnet with state-of-the-art methods on the orsi-4199 datasets.  $\uparrow$  means bigger is better and  $\downarrow$  means smaller is better. The top three results are marked in bold, italic, underline, respectively.

| Methods                                   | Year | ORSI-4199        |                            |                          |                |
|-------------------------------------------|------|------------------|----------------------------|--------------------------|----------------|
|                                           |      | MAE $\downarrow$ | $F_{\beta}^{max} \uparrow$ | $E_{\xi}^{max} \uparrow$ | $S_a \uparrow$ |
| MJRBNet <sup>†</sup> (Tu et al., 2021)    | 2021 | 0.0374           | 0.8493                     | 0.9311                   | 0.8593         |
| EMFINet <sup>†</sup> (Wang et al., 2022b) | 2022 | 0.0330           | 0.8584                     | 0.9340                   | 0.8675         |
| ERPNet <sup>†</sup> (Zhou et al., 2022)   | 2022 | 0.0357           | 0.8553                     | 0.9290                   | 0.8670         |
| HFAFNet <sup>†</sup> (Wang et al., 2022a) | 2022 | <u>0.0272</u>    | <u>0.8858</u>              | <u>0.9539</u>            | <b>0.8865</b>  |
| SeaNet <sup>†</sup> (Li et al., 2023c)    | 2023 | 0.0308           | 0.8653                     | 0.9426                   | 0.8722         |
| GLGCNet <sup>†</sup> (Bai et al., 2023)   | 2023 | 0.0274           | 0.8808                     | 0.9508                   | <u>0.8839</u>  |
| GeleNet <sup>†</sup> (Li et al., 2023b)   | 2023 | <i>0.0264</i>    | <i>0.8842</i>              | <i>0.9544</i>            | <i>0.8862</i>  |
| MCPNet <sup>†</sup> (Huang et al., 2024)  | 2024 | 0.0324           | 0.8667                     | 0.9402                   | 0.8736         |
| WeightNet <sup>†</sup> (Di et al., 2024)  | 2024 | 0.0293           | 0.8749                     | 0.9459                   | 0.8820         |
| PRNet <sup>†</sup> (Gu et al., 2024)      | 2024 | <b>0.0254</b>    | 0.8750                     | 0.9499                   | 0.8822         |
| BCARNet <sup>†</sup> (Gu et al., 2025)    | 2025 | 0.0306           | 0.8498                     | 0.9407                   | 0.8758         |
| LGIPNet <sup>†</sup> (Sun et al., 2025)   | 2025 | 0.0283           | 0.8769                     | 0.9483                   | 0.8833         |
| IEANet <sup>†</sup>                       | -    | <i>0.0264</i>    | <b>0.8889</b>              | <b>0.9566</b>            | <b>0.8865</b>  |

**Table 3**

Ablation study of our proposed modules. The best results are in.

| ID | Base | Modules |      |      | EORSSD           |                |                            |                          |
|----|------|---------|------|------|------------------|----------------|----------------------------|--------------------------|
|    |      | IEAM    | CAGM | SCPM | MAE $\downarrow$ | $S_a \uparrow$ | $F_{\beta}^{max} \uparrow$ | $E_{\xi}^{max} \uparrow$ |
| 1  | ✓    |         |      |      | 0.0087           | 0.9220         | 0.8698                     | 0.9760                   |
| 2  | ✓    | ✓       |      |      | 0.0057           | 0.9379         | 0.8864                     | 0.9784                   |
| 3  | ✓    |         | ✓    |      | 0.0060           | 0.9357         | 0.8870                     | 0.9785                   |
| 4  | ✓    |         |      | ✓    | 0.0061           | 0.9362         | 0.8868                     | 0.9798                   |
| 5  | ✓    | ✓       | ✓    |      | 0.0053           | 0.9393         | 0.8919                     | 0.9813                   |
| 6  | ✓    | ✓       |      | ✓    | 0.0050           | 0.9389         | 0.8923                     | 0.9819                   |
| 7  | ✓    |         | ✓    | ✓    | 0.0054           | 0.9382         | 0.8910                     | 0.9821                   |
| 8  | ✓    | ✓       | ✓    | ✓    | <b>0.0045</b>    | <b>0.9419</b>  | <b>0.8937</b>              | <b>0.9824</b>            |

sets of ORSSD, EORSSD, and ORSI-4199 are expanded to 4,800, 11,200, and 16,000 samples, respectively, and resize the images to  $384 \times 384$ . In addition, we use the parameters of the pre-trained UniFormer model (Li et al., 2023d) to initialize the parameters of the encoder network. Due to GPU memory constraints, the batch size is set to 8. The model is trained for 45 epochs using the Adam optimizer with a base learning rate of  $1 \times 10^{-4}$ , which follows a step-based decay strategy and is reduced by a factor of 0.1 every 30 epochs. The default Adam momentum parameters ( $\beta_1 = 0.9$ ,  $\beta_2 = 0.999$ ) are used without weight decay. Gradient clipping with a threshold of 0.5 is applied during training. All experiments are conducted in full FP32 precision. During inference, single-scale testing is performed without test-time augmentation.

#### 4.4. Comparison with state-of-the-art methods

##### 4.4.1. Comparison methods

We compared IEANet with 23 state-of-the-art saliency models on the ORSI-SOD test set, including LVNet (Li et al., 2019), DAFNet (Zhang et al., 2020), MJRBNet (Tu et al., 2021), EMFINet (Wang et al., 2022b), ERPNet (Zhou et al., 2022), HFAFNet (Wang et al., 2022a), SeaNet (Li et al., 2023c), GLGCNet (Bai et al., 2023), ASTTNet (Gao et al., 2023), GeleNet (Li et al., 2023b), ASNet (Yan et al., 2024), PRNet (Gu et al., 2024), ADSTNet (Zhao et al., 2024a), MCPNet (Huang et al., 2024), WeightNet (Di et al., 2024), MIRGNet (Zhao et al., 2024b), SFANet (Quan et al., 2024), GCRNet (Li et al., 2024b), BCARNet (Gu et al., 2025), and LGIPNet (Sun et al., 2025). For the sake of fairness and reproducibility, all saliency maps are produced using the official codebases provided by the original authors, and all performance metrics are obtained through standardized evaluation protocols commonly used in prior studies (Li et al., 2023c). It is worth noting that since some models only disclosed some of the performance indicators of ORSI-4199, we selected four representative indicators for comparison.

**Table 4**

Ablation experiments of IEAM with different branch combinations. Base represents the complete model excluding IEAM, the best results are in **bold**.

| ID | Model           | EORSSD           |                            |                          |                |
|----|-----------------|------------------|----------------------------|--------------------------|----------------|
|    |                 | MAE $\downarrow$ | $F_{\beta}^{max} \uparrow$ | $E_{\xi}^{max} \uparrow$ | $S_a \uparrow$ |
| 1  | Base            | 0.0054           | 0.8910                     | 0.9821                   | 0.9382         |
| 2  | Enh             | 0.0052           | 0.8924                     | 0.9814                   | 0.9399         |
| 3  | Enh + Edg       | 0.0050           | 0.8931                     | 0.9820                   | 0.9406         |
| 4  | Enh + Sem       | 0.0048           | 0.8945                     | 0.9829                   | 0.9411         |
| 5  | Enh + Edg + Sem | <b>0.0046</b>    | <b>0.8970</b>              | <b>0.9847</b>            | <b>0.9419</b>  |

**Table 5**

Ablation experiments of IEAM with different connectors. The best results are in **bold**.

| ID | Strategy | EORSSD           |                            |                          |                |
|----|----------|------------------|----------------------------|--------------------------|----------------|
|    |          | MAE $\downarrow$ | $F_{\beta}^{max} \uparrow$ | $E_{\xi}^{max} \uparrow$ | $S_a \uparrow$ |
| 1  | a        | 0.0057           | 0.8892                     | 0.9791                   | 0.9370         |
| 2  | b        | 0.0054           | 0.8894                     | 0.9792                   | 0.9381         |
| 3  | c        | 0.0048           | 0.8911                     | 0.9820                   | 0.9401         |
| 4  | d        | <b>0.0046</b>    | <b>0.8970</b>              | <b>0.9847</b>            | <b>0.9419</b>  |

#### 4.4.2. Quantitative comparison

As shown in Tables 1 and 2, our model outperforms most of the comparison models on all three datasets. For example, on the ORSSD and EORSSD datasets, except for the poor  $F_{\beta}^{adp}$  index, the other indices all achieved excellent performance, especially  $E_{\xi}^{max}$ , which scored 0.9872 and 0.9847 respectively. In ORSI-4199, we are in the lead in the other three indicators and also ranked second in the MAE indicator. Among the three datasets, ORSI-4199 is the most challenging because images usually contain multiple distractors as well as small salient objects. Besides, the precision-recall (PR) curve and F-measure curve shown in the Fig. 7 further illustrate that our method is always better than other methods overall. These results demonstrate that our framework is particularly effective in suppressing background noise and precisely localizing small salient targets even under complex scene conditions, which further highlights its practical value and versatility.

#### 4.4.3. Qualitative comparison

The Fig. 8 shows a qualitative visual comparison of different saliency models, including airplanes, cars, rivers, and lakes. Here, all test images are from the EORSSD dataset. According to the Fig. 8, we can find that the proposed IEANet outperforms other models, and the saliency map shown in the figure can accurately locate and segment salient objects in complex scenes. Specifically, we can see that many models fail to highlight salient objects. For example, the image in the second example contains multiple small cars, and some models have some problems with the integrity of the car. This shows that a good representation of salient objects is crucial for the SOD task. In particular, the position attention provides good localization and ensures the integrity of salient objects. In addition, when salient objects are in cluttered and complex scenes, the salient object boundaries predicted by other models become blurred, as shown in the 6th example. In contrast, due to the introduction of edge information, the position attention features of our model have precise boundary details, which helps to sharpen the salient map.

#### 4.5. Ablation study

In this section, we conduct ablation studies to evaluate the effectiveness of each component embedded in our model. Therefore, we analyze 1) the individual contributions of three modules, 2) the effectiveness of each component in IEAM, 3) the effectiveness of each component in CAGM, and 4) the effectiveness of each component in SCPM. We conducted these ablation studies on IEANet with UniFormer as the

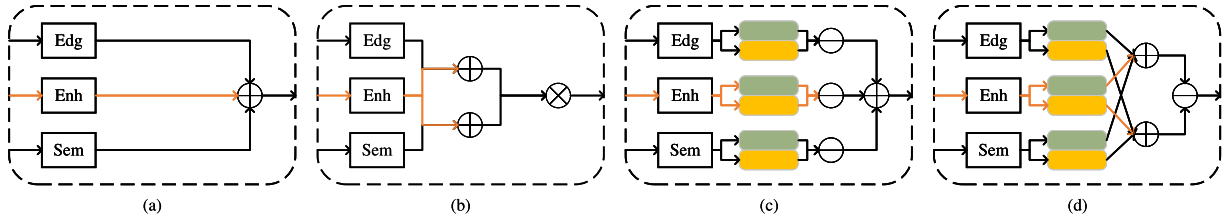


Fig. 10. IEAM with different connector construction. Edg, Enh, and Sem represent the three branches of IEAM.

Table 6

Ablation experiments on different components of CAGM. We conduct the quantitative comparison between the model Baseline, with (w/) RCA, with (w/) RSA, with (w/) TB and with (w/) CAGM. TB means Transformer block, the best results are in **bold**.

| ID | Schemes  | EORSSD        |                            |                          |                       |
|----|----------|---------------|----------------------------|--------------------------|-----------------------|
|    |          | MAE ↓         | $F_{\beta}^{max} \uparrow$ | $E_{\xi}^{max} \uparrow$ | $S_{\alpha} \uparrow$ |
| 1  | Baseline | 0.0087        | 0.8698                     | 0.9760                   | 0.9220                |
| 2  | w/ RCA   | 0.0072        | 0.8792                     | 0.9754                   | 0.9229                |
| 3  | w/ RSA   | 0.0070        | 0.8826                     | 0.9766                   | 0.9323                |
| 4  | w/ TB    | 0.0065        | 0.8862                     | 0.9779                   | 0.9341                |
| 5  | w/ CAGM  | <b>0.0060</b> | <b>0.8870</b>              | <b>0.9785</b>            | <b>0.9357</b>         |

Table 7

Ablation experiments of different depths of Transformer blocks. The best results are in **bold**.

| ID | Depth | EORSSD        |                            |                          |                       |
|----|-------|---------------|----------------------------|--------------------------|-----------------------|
|    |       | MAE ↓         | $F_{\beta}^{max} \uparrow$ | $E_{\xi}^{max} \uparrow$ | $S_{\alpha} \uparrow$ |
| 1  | 0     | 0.0052        | 0.8887                     | 0.9802                   | 0.9363                |
| 2  | 2     | 0.0049        | 0.8898                     | 0.9810                   | 0.9380                |
| 3  | 4     | <b>0.0045</b> | <b>0.8937</b>              | <b>0.9824</b>            | <b>0.9419</b>         |
| 4  | 6     | 0.0047        | 0.8922                     | 0.9820                   | 0.9382                |

backbone, and used the same parameter settings and dataset divisions as described above.

#### 4.5.1. Individual contributions of three modules.

To evaluate the individual contributions of the three proposed modules, including IEAM, CAGM, and SCPM, we design seven model variations by combining these modules in different configurations, 1) Baseline, which excludes all proposed modules, 2) Baseline + IEAM, 3) Baseline + CAGM, 4) Baseline + SCPM, 5) Baseline + IEAM + CAGM, 6) Baseline + IEAM + SCPM, and 7) Baseline + CAGM + SCPM. The quantitative results are presented in Table 3. To build a baseline model, we remove IEAM, CAGM, and SCPM from the final architecture of IEANet. Subsequently, these modules are gradually integrated into the baseline model, forming the second to fourth rows in the table. It is worth noting that the baseline model performs worse than most ORSI-SOD methods, while the performance steadily improves with the integration of key components. In particular, integrating only IEAM shows a significant improvement over the baseline model. Specifically, the integration of IEAM improves  $S_{\alpha}$  and MAE by 0.0101 and 0.0030, respectively. Moreover, compared with the model without IEAM, our full IEANet model improves  $S_{\alpha}$  and MAE by 0.0037 and 0.0009, respectively. These significant improvements highlight the necessity of using interactive edge perception in ORSI-SOD.

In addition, we conduct a more comprehensive ablation analysis by exploring different proposed combinations. Overall, the combined variants show significant improvements compared to the variants integrating only a single module. For example, the variant integrating both IEAM and CAGM outperforms the variants integrating only IEAM or CAGM. Similar observations are made in the sixth and seventh rows. Finally, as shown in the eighth row, the full IEANet achieves the high-

Table 8

Ablation experiments on different components of SCPM. We conduct the quantitative comparison between the model Baseline, with (w/) SP, with (w/) SA, and with (w/) SCPM. Where SP means semantic preservation, SA means self-attention, the best results are in **bold**.

| ID | Schemes  | EORSSD        |                            |                          |                       |
|----|----------|---------------|----------------------------|--------------------------|-----------------------|
|    |          | MAE ↓         | $F_{\beta}^{max} \uparrow$ | $E_{\xi}^{max} \uparrow$ | $S_{\alpha} \uparrow$ |
| 1  | Baseline | 0.0087        | 0.8698                     | 0.9760                   | 0.9220                |
| 2  | w/ SP    | 0.0075        | 0.8800                     | 0.9760                   | 0.9300                |
| 3  | w/ SA    | 0.0068        | 0.8823                     | 0.9762                   | 0.9313                |
| 4  | w/ SCPM  | <b>0.0061</b> | <b>0.8868</b>              | <b>0.9798</b>            | <b>0.9362</b>         |

est performance among all options, with improvements of 0.0141 in  $S_{\alpha}$  and 0.0042 in MAE on EORSSD. These improvements further verify the effectiveness of the proposed module in improving the performance of IEANet.

#### 4.5.2. Effectiveness of each component in IEAM

Table 4 shows the ablation experiments of IEAM with different branch combinations. Compared with the baseline, IEAM helps the model achieve excellent scores in all three datasets. As observed, the introduction of the Enh branch leads to a slight decrease in the E-measure, however, other evaluation metrics exhibit noticeable improvements. Moreover, the incorporation of additional auxiliary branches consistently demonstrates their respective contributions. The model equipped with all three branches achieves superior overall performance, underscoring the significance of the parallel three-branch architecture in IEAM.

To investigate the impact of different connection strategies on the IEAM module, we illustrate the evolution of our head designs in Fig. 10. Specifically, strategy (a) directly sums the three branches, treating edge, semantic, and enhanced features equally, but ignores their structural differences and limits effective feature complementarity. Strategy (b) introduces pairwise interactions via element-wise addition and multiplication, which moderately enhances salient responses; however, these interactions remain implicit and lack explicit structural modeling. In contrast, strategy (c) employs a pooling-based difference operation to explicitly model inter-branch feature discrepancies, strengthening edge-sensitive and discriminative representations. Strategy (d) further refines the pooling difference strategy by first aggregating the pooling results from all branches, and then applying a unified difference operation for feature fusion. The performance of each variant is summarized in Table 5. A clear performance improvement is observed when comparing strategies (a) and (c), indicating the effectiveness of the pooling difference mechanism in enhancing feature discrimination. These results validate that a carefully designed interaction among branches, combined with explicit edge-aware enhancement, plays a crucial role in maximizing the capacity of the IEAM.

#### 4.5.3. Effectiveness of each component in CAGM

Table 6 presents the ablation experiments of CAGM with different component configurations. In contrast, the integrated CAGM enables

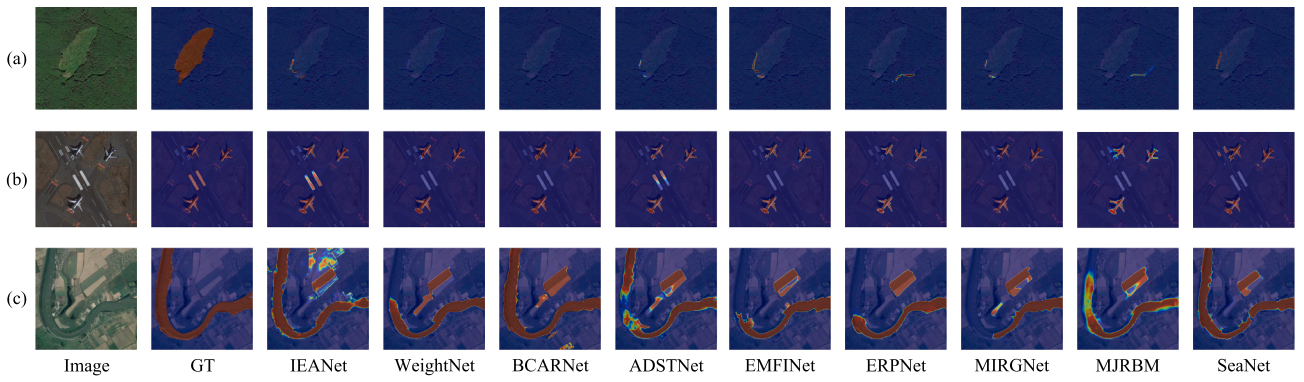


Fig. 11. Representative failure cases.

the model to achieve consistently superior performance on all four metrics. As observed, individually incorporating the RCA module, the RSA module, and the Transformer block each contributes to noticeable improvements over the baseline, demonstrating the effectiveness of these components in isolation. Furthermore, the model equipped with the complete CAGM configuration, which integrates RCA, RSA, and the Transformer block, achieves the best overall performance. These results validate the complementary strengths of the three modules and highlight the importance of their joint integration in enhancing feature interaction.

In addition, we conducted ablation experiments to investigate the effect of different Transformer block depths in Table 7. Specifically, four Transformer depths (0, 2, 4, and 6) were evaluated, where the Transformer depth is defined as the number of stacked Transformer blocks integrated into the network. When the Transformer depth is set to 0, the model relies solely on local feature interactions, which limits its ability to capture long-range contextual information. Increasing the depth from 0 to 4 layers continuously improves performance, indicating moderate self-attention effectively strengthens long-range dependency modeling and complements convolutional representations, and global context plays a key role in remote sensing salient object recognition. However, a further increase to six layers leads to performance degradation, as excessive depth causes feature redundancy and over-parameterization, weakening the generalization ability of the model. These findings demonstrate that a Transformer depth of 4 strikes an effective balance between representational capacity and model generalization, and thus is adopted as the default setting in our final CAGM design.

#### 4.5.4. Effectiveness of each component in SCPM

Table 8 presents the ablation results of the proposed SCPM. The progressive integration of its components consistently leads to improvements across all four evaluation metrics. To begin with, the introduction of the semantic prior (SP) yields significant performance gains by injecting category-aware cues into the decoding process, thereby enhancing semantic consistency, particularly in scenarios involving occlusion or deformation. In addition, incorporating the self-attention (SA) mechanism alone further improves performance by reinforcing the alignment between semantic and spatial features through global contextual modeling.

Notably, the full SCPM configuration integrating both SP and SA components demonstrates superior overall performance, thereby validating their complementary functions. When applied independently, SP primarily contributes to maintaining high-level semantic consistency during upsampling, while SA focuses on refining spatial details and enhancing boundary localization. In combination, these components effectively overcome critical challenges in ORSI-SOD, such as accurately identifying complex structures and suppressing background interference.

#### 4.6. Failure cases

Although our proposed model achieves significant performance improvements in the ORSI-SOD task, challenges remain in certain complex scenes, as illustrated by typical failure cases in Fig. 11. In case (a), the salient object exhibits highly similar colors to the background, making it difficult to distinguish foreground from background and resulting in the loss of salient regions. This limitation is likely due to insufficient discriminative power in the current multiscale feature fusion. In cases (b) and (c), incomplete extraction of salient objects and false detections of background areas occur, which can be attributed to imbalanced feature representation and limited edge sensitivity in the current architecture. These diagnostic observations motivate future work aimed at enhancing multiscale feature fusion and integrating stronger edge perception mechanisms to better capture complex structures, suppress background interference.

#### 5. Conclusion

In this paper, we propose IEANet for salient object detection in optical remote sensing images. The proposed method is designed as a unified framework that emphasizes coordinated representation learning across multiple feature levels. Within this framework, edge-aware representations, global-local attention mechanisms, and semantic-preserving reconstruction are jointly incorporated to alleviate boundary ambiguity and semantic inconsistency. Specifically, edge-aware representations are enhanced through interactive feature perception to refine object boundaries, global-local attention mechanisms are employed to jointly model fine-grained details and long-range contextual dependencies, and semantic-preserving reconstruction is introduced to compensate for semantic degradation during upsampling and maintain multiscale feature consistency. Extensive experiments on three challenging datasets demonstrate the superiority and effectiveness of the proposed method. In future research, efforts will focus on enhancing the effectiveness of feature interaction and representation learning to better address camouflaged targets. Meanwhile, more sophisticated multi-scale feature fusion mechanisms will be developed to better capture complex structures and handle camouflaged or low-contrast objects in challenging remote sensing scenarios. In addition, improving the quality and diversity of training data remains an important direction for further strengthening model robustness and generalization.

#### CRediT authorship contribution statement

**Xian Fang:** Conceptualization, Supervision, Writing – review and editing, Funding acquisition. **Qihang Wang:** Methodology, Writing – original draft, Software. **Qiaohong Chen:** Writing – review and editing. **Gongyang Li:** Visualization, Investigation, Data preparation, Funding acquisition.



## Data availability

Data will be made available on request.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Supplementary material

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